

Conditional Information Bottleneck Approach For Time Series Imputation

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1. Introduction

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CONDITIONAL INFORMATION BOTTLENECK APPROACH FOR TIME SERIES IMPUTATION

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ABSTRACT

Time series imputation presents a significant challenge because it requires capturing the underlying temporal dynamics from partially observed time series data. Among the recent successes of imputation methods based on generative models, the *information bottleneck* (IB) framework offers a well-suited theoretical foundation for multiple imputations, allowing us to account for the uncertainty associated with the imputed values. However, direct application of IB framework to time series data without considering their temporal context can lead to a substantial loss of temporal dependencies, which, in turn, can degrade the overall imputation performance. To address such a challenge, we propose a novel *conditional information bottleneck* (CIB) approach for time series imputation, which aims to mitigate the potentially negative consequences of the regularization constraint by focusing on reducing the redundant information conditioned on the temporal context. We provide a theoretical analysis of its effect by adapting variational decomposition. We use the resulting insight and propose a novel deep learning method that can approximately achieve the proposed CIB objective for time series imputation as a combination of evidence lower bound and novel temporal kernel-enhanced contrastive optimization. Our experiments, conducted on multiple real-world datasets, consistently demonstrate that our method significantly improves imputation performance (including both interpolation and extrapolation), and also enhances prediction performance based on the imputed values.

1 INTRODUCTION

Multivariate time series data often includes missing features, with diverse missing ratios and patterns depending on distinct sampling periods or measurement strategies (Johnson et al., 2016). Since these missing features can significantly impair the performance of downstream tasks and comprehension of the temporal dynamics, time series imputation, which aims to reconstruct the missing features, has become a pivotal and pervasive topic across numerous practical domains, including healthcare, environmental science, and various other fields. What makes time series imputation challenging is that an imputation method must satisfy two essential requirements: i) it must account for underlying temporal dependencies, and ii) it should allow for *multiple imputations* to facilitate uncertainty quantification for real-world decision-making.

Generative models, particularly variational autoencoders (VAEs) (Kingma & Welling, 2014), have been employed in the context of multiple imputation tasks due to their capability to generate samples in a probabilistic manner. VAE-based imputation methods primarily focus on defining the evidence lower bound, where the reconstruction error is computed only over the observed part of the incomplete data (Sohn et al., 2015; Nazabal et al., 2020). These methods can be naturally interpreted under the information bottleneck (IB) principle (Tishby & Zaslavsky, 2015), providing an information-theoretic understanding of what constitutes an imputation-relevant representation. This understanding is based on the fundamental trade-off between maintaining a concise representation (i.e., regularization) and preserving good representation power (i.e., reconstruction) (Voloshynovskiy et al., 2019).

However, a direct application of the IB principle to time series imputation struggles with capturing the underlying temporal dependencies. Our motivating examples in Figure 1(B) show that imputation methods trained with the conventional IB framework lose a significant amount of information about temporal dynamics relevant for imputing missing values. In this paper, we theoretically analyze that

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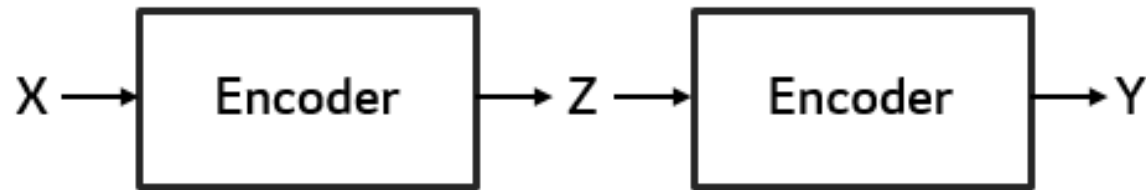
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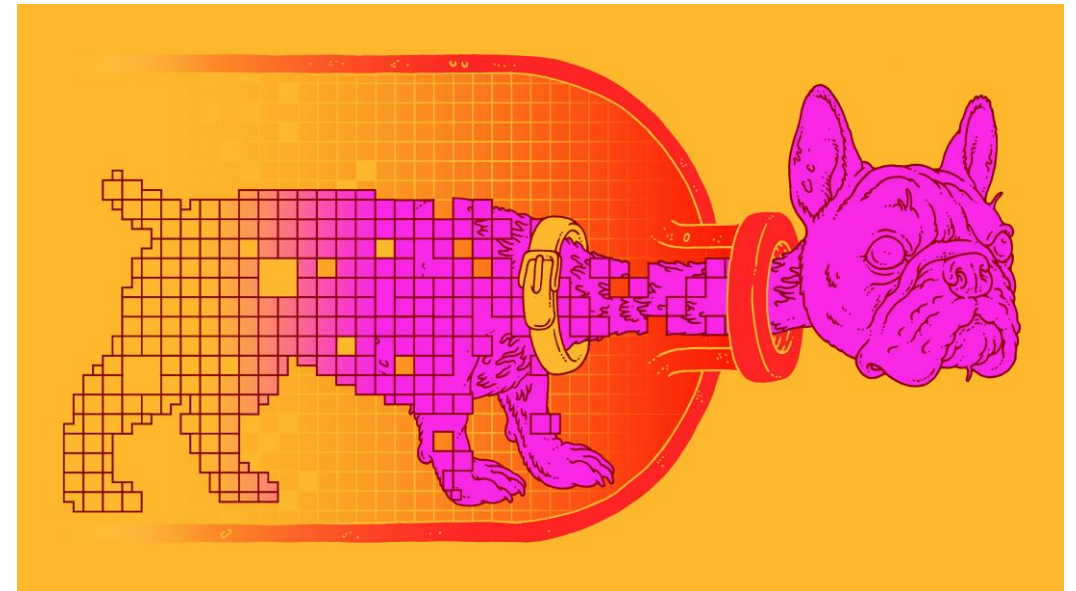
2. Introduction

What is Information Bottleneck(IB)?

- 복잡한 신호나 데이터에서 중요한 정보를 필터링하는 방법(Framework)
- 다양한 방법론들에 적용 가능!
- 필요한 정보만 캐내어 원하는 목적을 “**효율적**” 으로 이루고하자는 욕구의 데이터과학적 실현



General Deep Learning Framework



2. Introduction

What is Information Bottleneck(IB)?

- **Entropy** : 데이터 X에 대해 기대할 수 있는 정보량/불확실성

$$\text{Entropy} : H(X) = E_{p(X)}[-\log p(X)] = \sum_x (-\log p(x)) p(x)$$

- **Conditional Entropy** : Y가 주어졌을 때 X에 기대할 수 있는 정보량

$$H(X|Y) = E_{p(X,Y)}[-\log p(X|Y)] = H(X,Y) - H(Y)$$

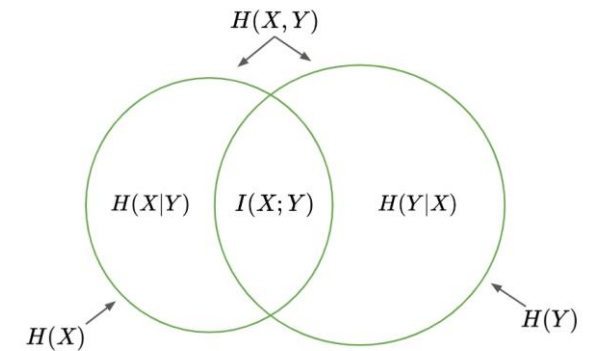
- **Relative entropy**(= Kullback-Leibler (KL) distance)

$$D_{KL}(p||q) = E_p \left[\log \frac{p(X)}{q(X)} \right]$$

- **Mutual Information**

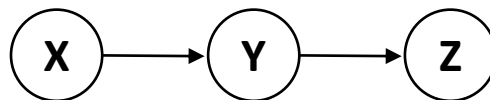
$$I(X;Y) = D_{KL}(p(x,y)||p(x)p(y)) = H(X) - H(X|Y)$$

- X와 Y의 통계적 의존성이 클수록 두 분포간의 차이가 커짐



2. Introduction

What is Information Bottleneck(IB)?



Markov chain

- Data-preprocessing inequality(DPI)

$$I(X; Y) \geq i(X; Z)$$

- DPI corollary

$$I(X; Y) \geq I(X; f(Y))$$

IB principle

$$\min_{\phi, \theta} I(X; Z) = I(X; Z) - \beta I(\theta; Z) \quad s.t. \quad \beta \geq 0$$

Lagrange multipliers

관심 있는 정보

- β 가 크면 클수록 $I(X; Z)$ 를 굳이 X 를 Compress할 필요가 없음($Z=X$ 일때가 $I(\theta; Z)$ 가 최대)
- β 가 0이면 $I(X; Z)$ 를 최소화하는 문제, Z 에 θ 의 정보 X

Conclusion

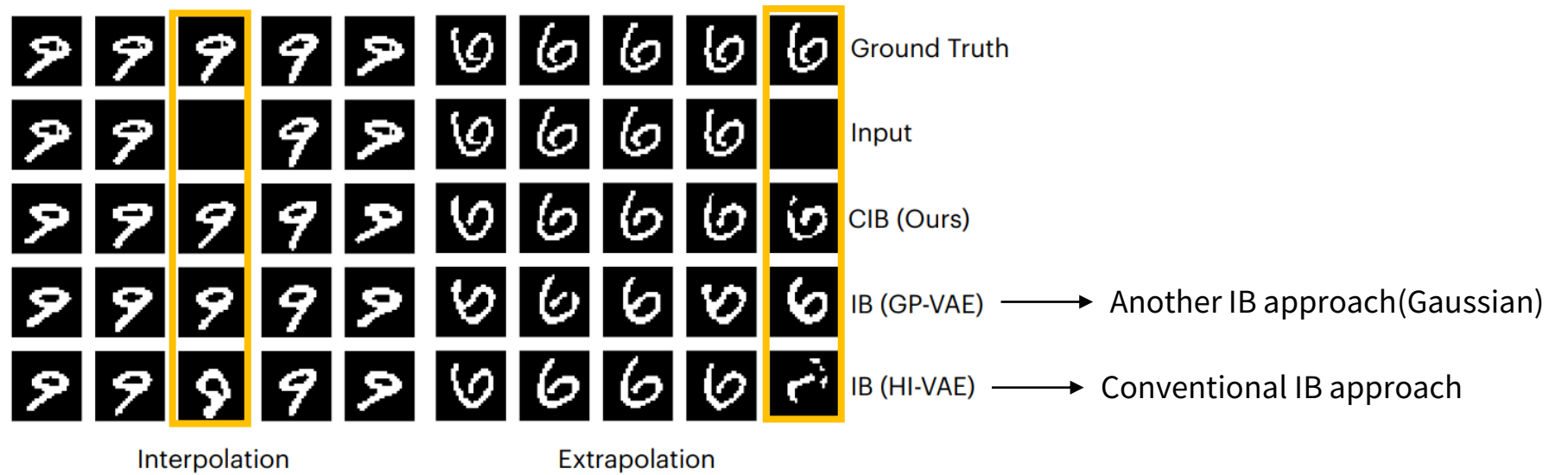
- IB :관심 있는 θ 에 대한 정보를 얻기 위해 데이터 X 를 가공 -> General한 모형에도 정보 손실량을 최소화하는 framework

2. Introduction

Time series imputation challenging

- Account for underlying **temporal dependencies**
- Allow for multiple imputations to facilitate **uncertainty quantification** for real-world decision-making

Motivation



3. Method

Definition 1 (Imputation)

- X^o : random variables for partially observed features
- X^m : missing features of X
- $X = X^o \cup X^m$

Unsupervised IB

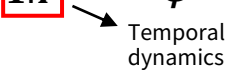
$$\min_{\phi, \theta} I(Z; X^o) = I_{\phi}(Z; X^o) - \beta I_{\theta}(X; Z)$$

ϕ, θ : learnable parameters

Objective(*Multiple imputation*)

- x_t^o : Observed features at each time step
- x_t^m : Missing features at each time step
- $x_t = x_t^o \cup x_t^m$

$$\min_{\phi, \theta} I_{\phi}(Z_t; X_{1:T}^o) = I_{\phi}(Z_t; X_{1:T}^o) - \beta I_{\theta}(X_t; Z_t) \longrightarrow \text{e.g., RNN or Transformer}$$

Temporal dynamics

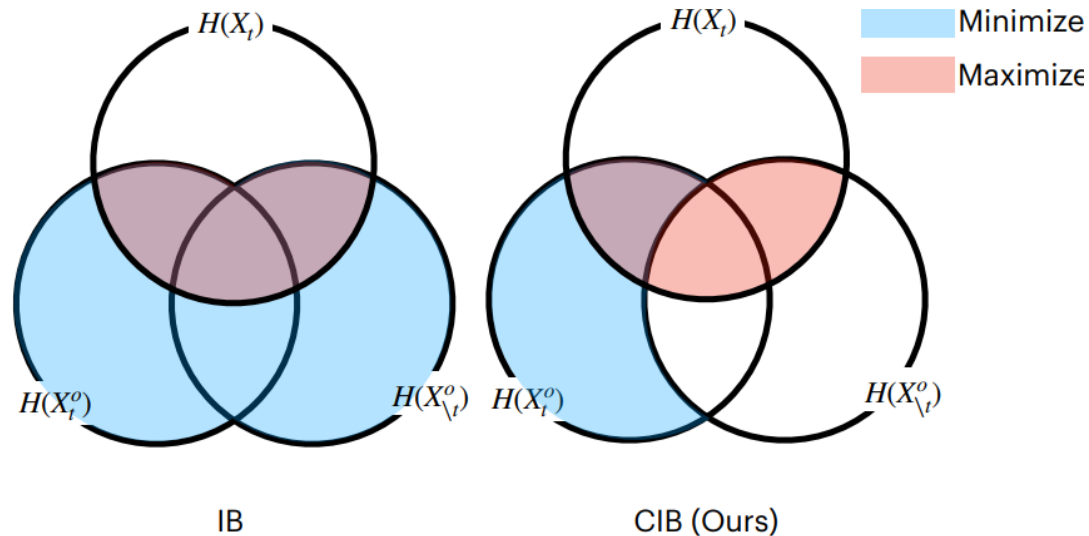
$\longrightarrow$ Loss of information(rely on particular time point(x_t^m))

3. Method

Definition 2 (Time Series Imputation)

- $X_{\setminus t}^o$: represents the random variables for the remaining input observations, excluding X_t^o

$$\min_{\phi, \theta} \underbrace{I_{\phi}(Z_t; X_t^o | X_{\setminus t}^o)}_{\text{Conditional Regularization}} - \underbrace{\beta I_{\theta}(X_t; Z_t)}_{\text{Reconstruction}}$$



3. Method

Deep Variational Conditional Information Bottleneck On Time Series

$$\min_{\phi, \theta} \underbrace{I_{\phi}(Z_t; X_t^o | X_{\setminus t}^o)}_{\text{Conditional Regularization}} - \underbrace{\beta I_{\theta}(X_t; Z_t)}_{\text{Reconstruction}} \rightarrow \text{Variational decomposition (단순한 분포로 근사(ELBO))}$$

Maximizing Reconstruction

$$\begin{aligned} I_{\theta}(\mathbf{X}_t; \mathbf{Z}_t) &= \mathcal{H}(\mathbf{x}_t) + D_{\text{KL}}(p(\mathbf{x}_t | \mathbf{z}_t) || p_{\theta}(\mathbf{x}_t | \mathbf{z}_t)) + \mathbb{E}_{\mathbf{x}_{1:T}^o \sim p_{\text{data}}} \left[\mathbb{E}_{\mathbf{z}_t \sim q_{\phi}(\mathbf{z}_t | \mathbf{x}_{1:T}^o)} [\log p_{\theta}(\mathbf{x}_t | \mathbf{z}_t)] \right] \\ &\geq \mathbb{E}_{\mathbf{x}_{1:T}^o \sim p_{\text{data}}} \left[\mathbb{E}_{\mathbf{z}_t \sim q_{\phi}(\mathbf{z}_t | \mathbf{x}_{1:T}^o)} [\log p_{\theta}(\mathbf{x}_t | \mathbf{z}_t)] \right] \stackrel{\text{def}}{=} -\mathcal{L}_{\phi, \theta}^1 \quad \text{Lower Bound} \end{aligned}$$

$p_{\theta}(X_t | Z_t) = N(\mu_{\theta}(Z_t), \text{diag}(\sigma_{\theta}(Z_t))) \rightarrow$ isotropic Gaussian으로 추정됨

3. Method

Deep Variational Conditional Information Bottleneck On Time Series

$$\min_{\phi, \theta} \underbrace{I_{\phi} \left(\mathbf{Z}_t; \mathbf{X}_t^o \mid \mathbf{X}_{\setminus t}^o \right)}_{\text{Conditional Regularization}} - \underbrace{\beta I_{\theta}(\mathbf{X}_t; \mathbf{Z}_t)}_{\text{Reconstruction}}$$

$$I(\mathbf{X}_t; \mathbf{Z}_t \mid \mathbf{X}_{\setminus t}^o) = I(\mathbf{X}_t; \mathbf{Z}_t, \mathbf{X}_{\setminus t}^o) - I(\mathbf{X}_t; \mathbf{X}_{\setminus t}^o).$$

Target($\mathbf{X}_t$)에 대한 정보량을 잃게 함

3. Method

Deep Variational Conditional Information Bottleneck On Time Series

$$\min_{\phi, \theta} \underbrace{I_{\phi}(Z_t; X_t^o | X_{\setminus t}^o)}_{\text{Conditional Regularization}} - \underbrace{\beta I_{\theta}(X_t; Z_t)}_{\text{Reconstruction}}$$

Minimizing Conditional Regularization

$$\min_{\phi, \theta} I(Z_t; X_t^o | X_{\setminus t}^o) \xrightarrow{\text{Chain rule}} \min_{\phi, \theta} I(Z_t; X_{1:T}^o) - I_{\theta}(Z_t; X_{\setminus t}^o)$$

- Encourage the latent representation to be concise
- Capture the underlying temporal dynamics

3. Method

Deep Variational Conditional Information Bottleneck On Time Series

Minimizing Conditional Regularization

$$\min_{\phi, \theta} I(\mathbf{Z}_t; \mathbf{X}_t^o | \mathbf{X}_{\setminus t}^o) = \min_{\phi, \theta} I(\mathbf{Z}_t; \mathbf{X}_{1:T}^o) - I_{\theta}(\mathbf{Z}_t; \mathbf{X}_{\setminus t}^o)$$

$$I(\mathbf{Z}_t; \mathbf{X}_{1:T}^o) \leq \mathbb{E}_{\mathbf{x}_{1:T}^o \sim p_{\text{data}}} [D_{\text{KL}}(q_{\phi}(\mathbf{z}_t | \mathbf{x}_{1:T}^o) || p(\mathbf{z}_t))] \stackrel{\text{def}}{=} \mathcal{L}_{\phi}^2$$

Upper Bound

$q_{\phi}(\mathbf{z}_t; \mathbf{X}_{1:T}^o) = N(\mu_{\theta}(\mathbf{X}_{1:T}^o), \text{diag}(\sigma_{\phi}(\mathbf{X}_{1:T}^o))) \rightarrow$ Stochastic encoder(Multivariate Gaussian)

$$I(\mathbf{Z}_t; \mathbf{X}_{\setminus t}^o) \geq \mathbb{E}_{\mathbf{x}_{1:T}^o \sim p_{\text{data}}} \left[\log \left(\frac{\sum_{t' \in \{1, \dots, T\} \setminus t} \exp(\mathbf{z}_t^T \tilde{\mathbf{z}}_{t'} / \tau)}{\sum_{\substack{\mathbf{x}_{1:T}^- \in \mathcal{X}_{1:T}^- \\ t' \in \{1, \dots, T\}, \\ \mathbf{z}_{t'}^- \sim q_{\phi}(\mathbf{z}_{t'}^- | \mathbf{x}_{1:T}^-)}} \exp(\mathbf{z}_t^T \mathbf{z}_{t'}^- / \tau)} \right) \right] \stackrel{\text{def}}{=} -\mathcal{L}_{\phi}^3$$

InfoNCE minimization
(contrastive learning)

Lower Bound – contrastive learning

3. Method

Deep Variational Conditional Information Bottleneck On Time Series

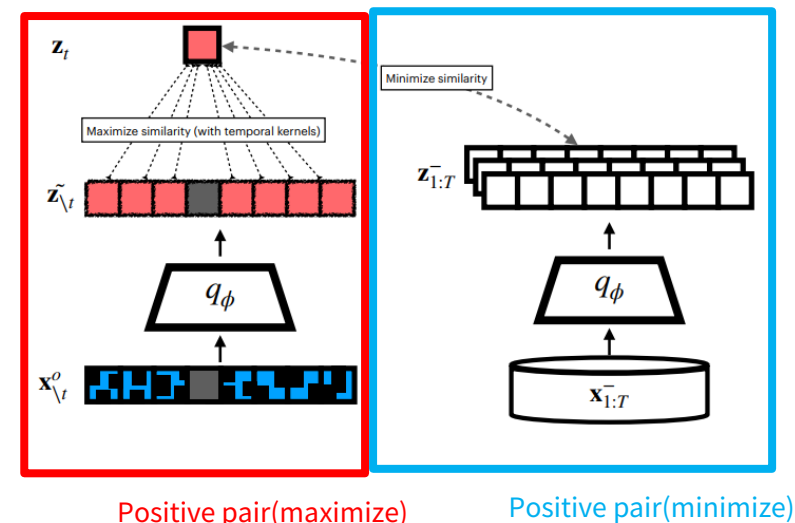
$$I(\mathbf{Z}_t; \mathbf{X}_{\setminus t}^o) \geq \mathbb{E}_{\mathbf{x}_{1:T}^o \sim p_{\text{data}}} \left[\log \left(\frac{\sum_{t' \in \{1, \dots, T\} \setminus t} \exp(\mathbf{z}_t^T \tilde{\mathbf{z}}_{t'} / \tau)}{\sum_{\mathbf{x}_{1:T}^- \in \mathcal{X}_{1:T}^-} \sum_{\substack{t' \in \{1, \dots, T\}, \\ \mathbf{z}_{t'}^- \sim q_\phi(\mathbf{z}_{t'}^- | \mathbf{x}_{1:T}^-)}} \exp(\mathbf{z}_t^T \mathbf{z}_{t'}^- / \tau)} \right) \right] \stackrel{\text{def}}{=} -\mathcal{L}_\phi^3$$

Kernel coefficient function

$$c_{\text{cauchy}}(\tau, \tau') = \sigma^2 \left(1 + \frac{(\tau - \tau')^2}{l^2} \right)^{-1}, \quad c_{\text{periodic}}(\tau, \tau') = \sigma^2 \exp \left(-\frac{2 \sin^2(\pi(\tau - \tau')/p)}{l^2} \right)$$

- **Cauchy kernel** : 두 데이터 포인트의 간의 거리를 정규화함, 노이즈에 덜 민감
- **Periodic kernel** : 데이터의 주기적 패턴 파악에 용이

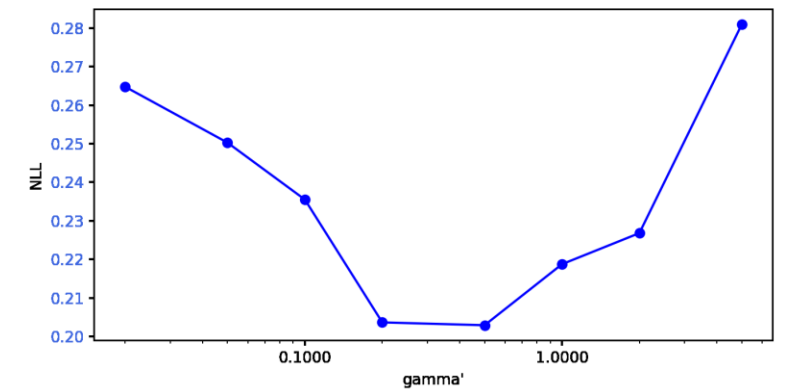
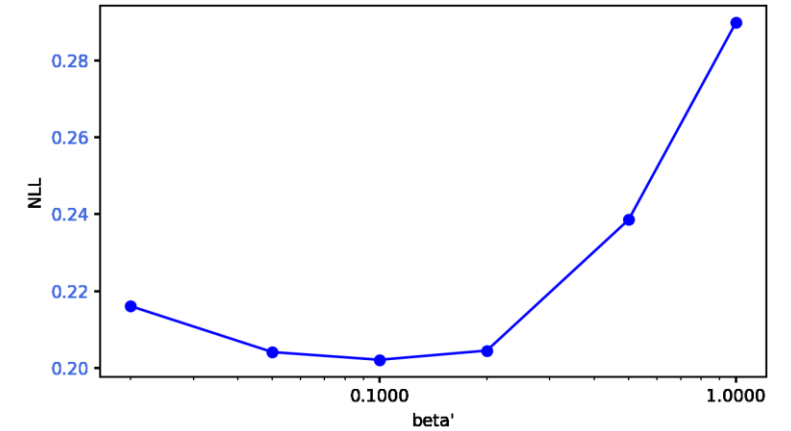
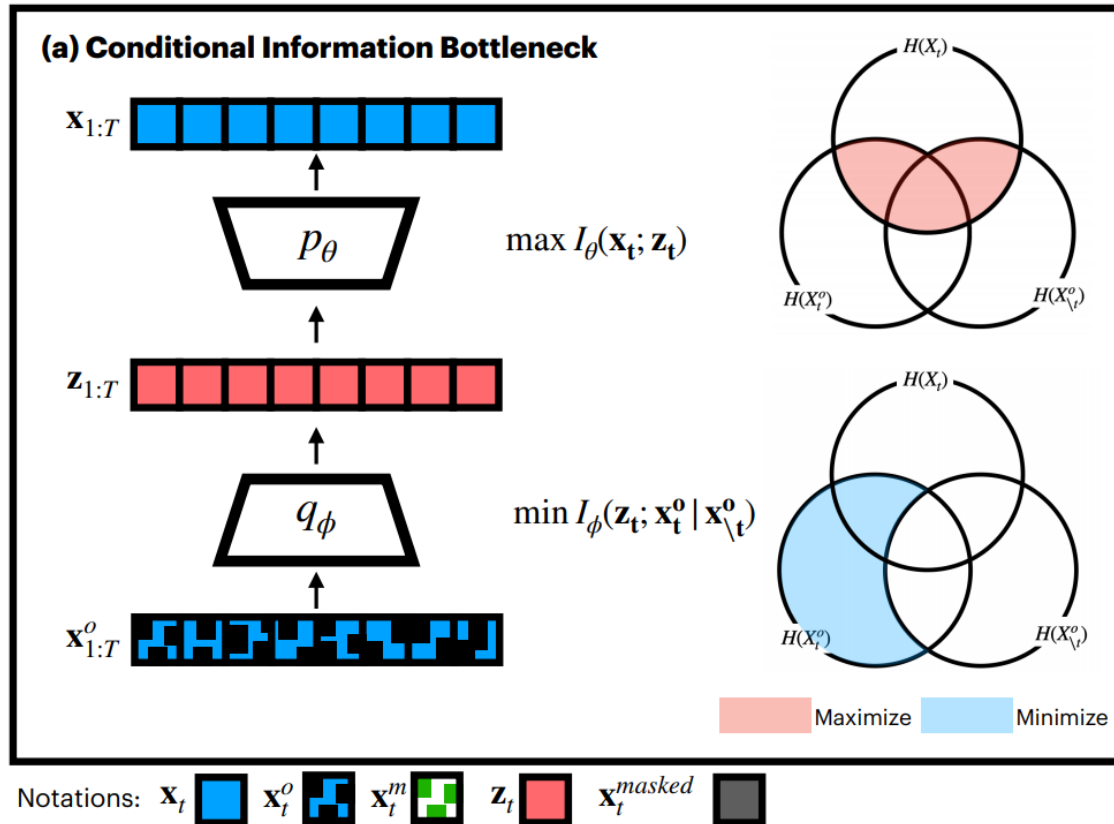
(b) Calculation of $I_\phi(\mathbf{z}_t; \mathbf{x}_{\setminus t}^o)$



3. Method

Optimization

$$\min_{\phi, \theta} \beta \mathcal{L}_{\phi, \theta}^1 + \mathcal{L}_{\phi}^2 + \gamma \mathcal{L}_{\phi}^3$$



4. Experiments

Measures feature-wise reconstruction

- NLL(Negative log-likelihood and MSE(Means squared error)
- Imputed values on artificially missing features

Prediction performance

- Imputed values to predict the target labels
- AUROC, MSE

Time series Imputation

- Predictive Methods
M-RNN, BRITS(2018) : RNN-based
SATIS(2023) -> self-attention mechanism -> Single imputation(Uncertainty)
- Generative Methods
GP-VAE : Autoencoder architecture
CSDI : diffusion-based approach
mTANs : attention-based autoencoder approach

Table C2: Number of learnable parameters.

	Ours	VAE	HI-VAE	GP-VAE	BRITS	SAITS	mTANs	CSDI
HealingMNIST	1.2M	1.2M	1.2M	1.2M	-	-	-	-
RotatedMNIST	1.2M	1.2M	1.2M	1.2M	-	-	-	-
Beijing	368K	368K	368K	368K	343K	362K	398K	375K
US Local	68K	68K	68K	68K	51K	87K	63K	62K
Physionet2012	21K	21K	21K	21K	24K	27K	29K	33K

4. Experiments

Imputation on Image Sequences

HealingMNIST : Time step 마다 60% MNAR로 결측된 데이터, 흰픽셀의 결측 확률이 검은 픽셀의 두배

RotatedMNIST : 랜덤한 결측을 가지는 데이터셋, Time dependency를 기반으로 결측 값을 재구성해야함

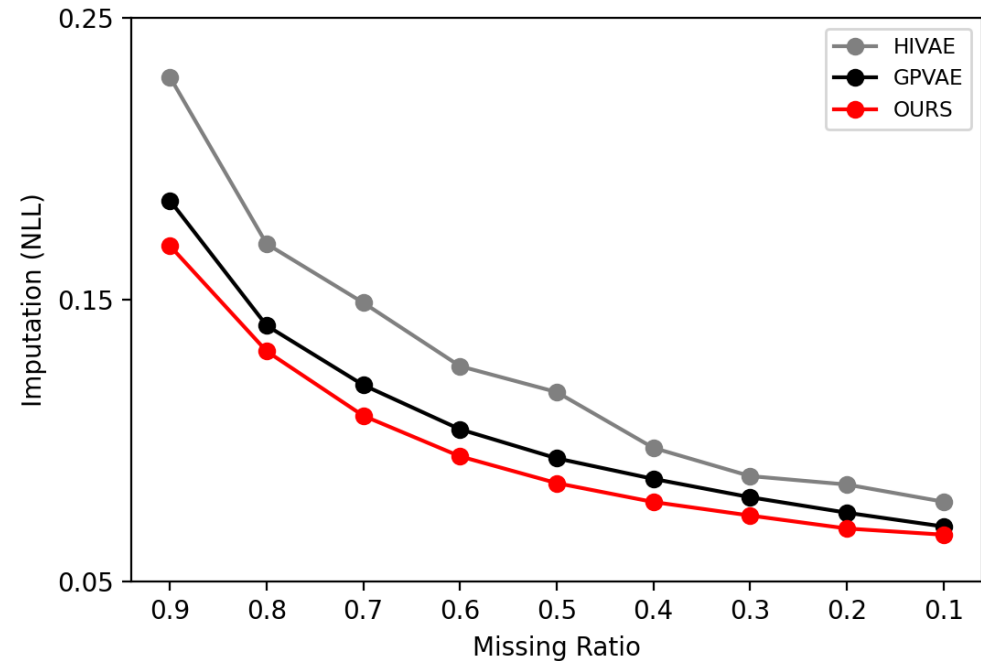
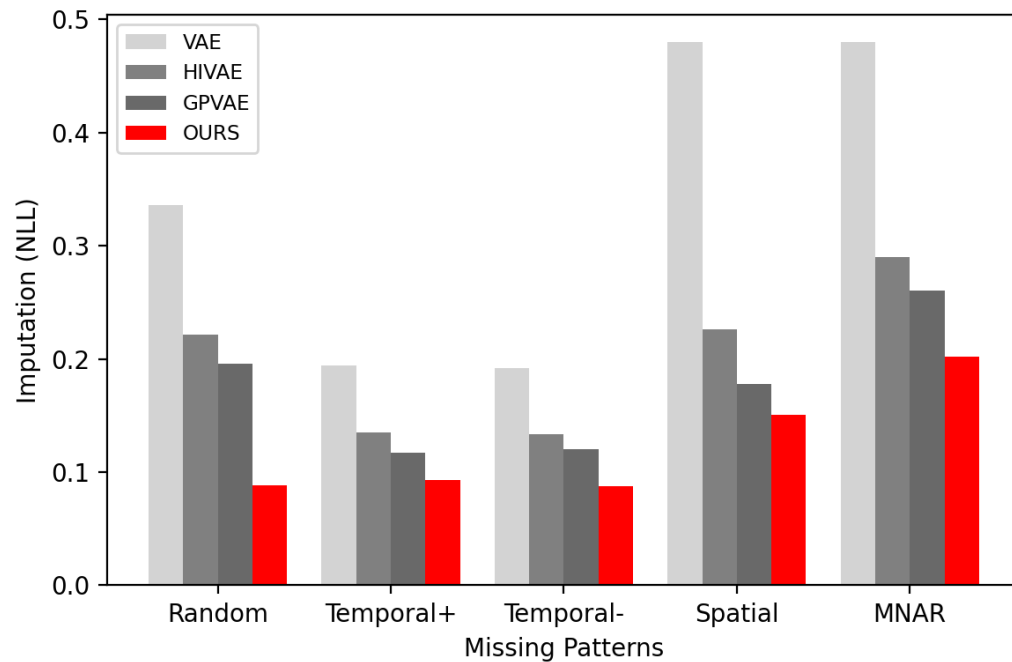


Methods	HealingMNIST (<i>missing with MNAR pattern</i>)			RotatedMNIST (<i>interpolation & extrapolation</i>)	
	NLL(↓)	MSE(↓)	AUROC(↑)	NLL(↓)	MSE(↓)
No Imp.	-	0.293 ± 0.000	0.920 ± 0.000	-	0.133 ± 0.000
Mean Imp.	-	0.168 ± 0.000	0.938 ± 0.000	-	0.085 ± 0.000
Forward Imp.	-	0.177 ± 0.000	0.946 ± 0.000	-	0.080 ± 0.000
VAE	0.480 ± 0.002	0.232 ± 0.000	0.922 ± 0.000	1.773 ± 0.127	0.133 ± 0.000
HI-VAE	0.290 ± 0.001	0.134 ± 0.003	0.962 ± 0.001	0.207 ± 0.007	0.087 ± 0.001
GP-VAE	0.261 ± 0.001	0.114 ± 0.002	0.960 ± 0.002	0.190 ± 0.001	0.080 ± 0.004
Ours(Uniform)	0.204 ± 0.002	0.090 ± 0.001	0.967 ± 0.001	0.184 ± 0.001	0.077 ± 0.001
Ours(Cauchy)	0.202 ± 0.004	0.088 ± 0.002	0.967 ± 0.000	0.184 ± 0.001	0.076 ± 0.002

4. Experiments

Imputation on Image Sequences

- Missing Patterns : Random, Temporal +/- (positive/negative temporal correlation), Spatial(neighboring pixels have correlate missing prob)
- Missing Ratio : 10% ~ 90% with random missing pattern



HealingMNIST

4. Experiments

Imputation for weather forecasting

- Beijing air Quality, US LOCAL : time series measurements are collected every hour.
- Temporal kernels(**Periodic kernel**)

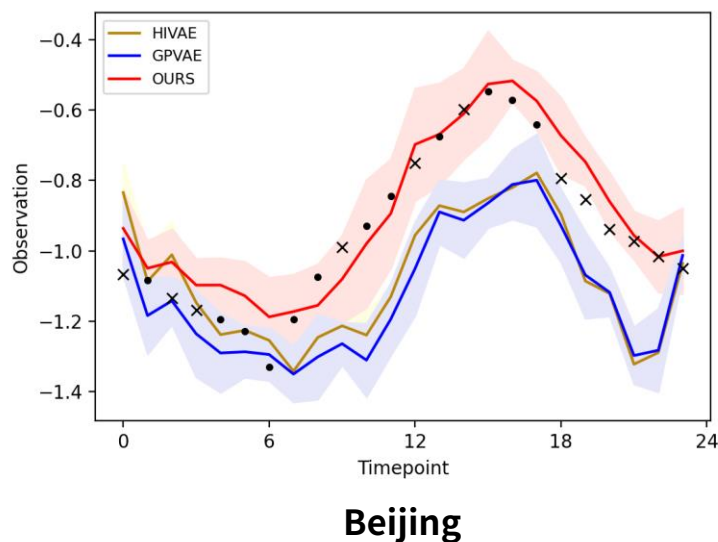
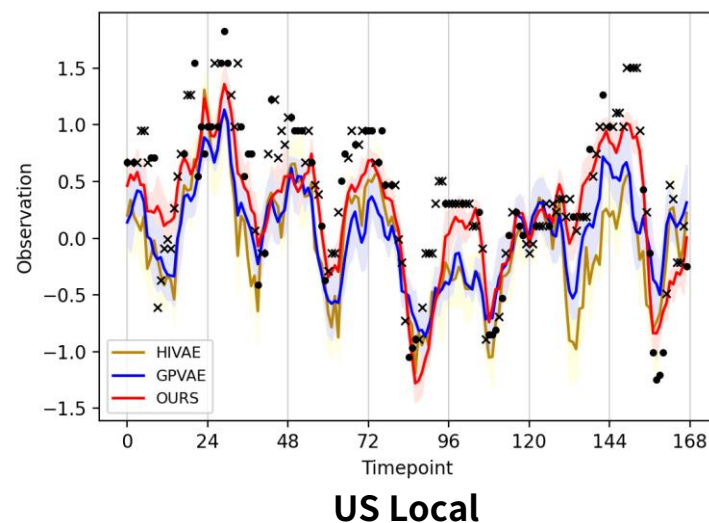


Table 2: Imputation and prediction performance on the weather forecasting datasets.

Methods	Beijing ($T=24$)			US Local ($T=168$)		
	NLL($\downarrow$)	MSE($\downarrow$)	ForecastMSE($\downarrow$)	NLL($\downarrow$)	MSE($\downarrow$)	ForecastMSE($\downarrow$)
No Imp.	-	1.015 ± 0.000	0.539 ± 0.000	-	1.113 ± 0.000	0.610 ± 0.000
Mean Imp.	-	0.460 ± 0.000	0.517 ± 0.000	-	0.509 ± 0.000	0.432 ± 0.000
Forward Imp.	-	0.399 ± 0.000	0.502 ± 0.000	-	0.391 ± 0.000	0.401 ± 0.000
BRITS	-	0.396 ± 0.002	0.490 ± 0.005	-	0.384 ± 0.001	0.398 ± 0.027
SAITS	-	0.283 ± 0.013	0.450 ± 0.005	-	0.275 ± 0.002	0.350 ± 0.067
mTANs	*	0.287 ± 0.005	0.436 ± 0.005	*	0.268 ± 0.018	0.337 ± 0.033
CSDI($n=5$)	*	0.287 ± 0.003	0.423 ± 0.003	*	0.378 ± 0.001	0.364 ± 0.036
CSDI($n=25$)	*	0.270 ± 0.001	0.423 ± 0.006	*	0.340 ± 0.000	0.347 ± 0.036
VAE	1.427 ± 0.001	1.016 ± 0.002	0.524 ± 0.006	1.462 ± 0.002	1.086 ± 0.004	0.467 ± 0.034
HI-VAE	1.081 ± 0.003	0.321 ± 0.008	0.464 ± 0.008	1.078 ± 0.005	0.317 ± 0.010	0.380 ± 0.060
GP-VAE	1.077 ± 0.006	0.316 ± 0.011	0.463 ± 0.008	1.078 ± 0.005	0.318 ± 0.010	0.385 ± 0.051
Ours(Uniform)	1.063 ± 0.001	0.291 ± 0.004	0.445 ± 0.003	1.052 ± 0.001	0.265 ± 0.002	0.351 ± 0.060
Ours(Periodic)	1.060 ± 0.002	0.283 ± 0.004	0.443 ± 0.004	1.049 ± 0.002	0.260 ± 0.003	0.327 ± 0.022



4. Experiments

Imputation for electrical health records

Physionet2012 – Mortality Prediction Challenge
ICU 환자의 48시간 병원 기록(환자 사망 예측)
Missing ratio : 80%

Table 3: Imputation and prediction performance on the clinical dataset.

Methods	Physionet2012 (<i>mortality prediction</i>)		
	NLL(↓)	MSE(↓)	AUROC(↑)
No Imp.	-	0.962 ± 0.000	0.692 ± 0.000
Mean Imp.	-	0.511 ± 0.000	0.703 ± 0.000
Forward Imp.	-	0.613 ± 0.000	0.710 ± 0.000
BRITS	-	0.529 ± 0.004	0.700 ± 0.005
SAITS	-	0.501 ± 0.024	0.713 ± 0.007
mTANs	*	0.499 ± 0.008	0.721 ± 0.004
CSDI(n=5)	*	0.548 ± 0.014	0.705 ± 0.005
CSDI(n=25)	*	0.478 ± 0.002	0.683 ± 0.033
VAE	1.400 ± 0.000	0.962 ± 0.000	0.691 ± 0.001
HI-VAE	1.345 ± 0.009	0.852 ± 0.018	0.696 ± 0.004
GP-VAE	1.227 ± 0.007	0.616 ± 0.013	0.730 ± 0.006
Ours(Uniform)	1.183 ± 0.007	0.528 ± 0.014	0.744 ± 0.009
Ours(Cauchy)	1.179 ± 0.006	0.521 ± 0.012	0.744 ± 0.009

5. Conclusion

Key idea

- Conditional IB : capture temporal dependency
- Imputation 방법론으로 인한 예측 성능 향상 입증

Pros

- 다양한 시계열 데이터셋에서 모두 우수한 성능을 보임
- 다양한 모델에서의 활용 가능성

Limitation

- 대용량 데이터셋에서의 검증을 하지 않음
- 계산 복잡도 및 시간 비용에서의 한계가 존재함

THANK YOU
